

Use Case Descriptions

Digital Wallet System — DigiWallet

Use Case 1: Create Account

Use Case Name	Create Account
Actor(s):	User
Description:	Allows a new user to register an account by providing personal information and credentials.
Precondition:	User does not have an existing account in the system.

Step #	Actor	System
1	User selects "Create Account" and enters name, email, phone, and password.	System validates that all fields are filled.
2		System saves the account to the users file with a starting balance of \$0.00.
3	User selects "Log In" or "Exit".	System navigates accordingly.

Alternate Courses:	If any field is empty, the account is not created and an error is displayed.
Implementation Constraints:	Email must be unique. All fields are required.

Use Case 2: Log In

Use Case Name	Log In
Actor(s):	User
Description:	Allows a registered user to authenticate and access their wallet.
Precondition:	User has an existing account registered in the system.

Step #	Actor	System
1	User enters their email and password.	System searches the users file for matching credentials.
2		System loads the user's wallet balance, transaction history, and linked payment methods.
3		System opens the main wallet menu for the authenticated user.

Alternate Courses:	If credentials are invalid, the system displays an error message. If no accounts exist, the system informs the user.
Implementation Constraints:	Email and password are required. Password is case-sensitive.

Use Case 3: Send Money

Use Case Name	Send Money
Actor(s):	User
Description:	Allows the authenticated user to transfer funds from their wallet balance to another person.
Precondition:	User is logged in and has a sufficient wallet balance.

Step #	Actor	System
1	User selects "Send Money" and enters the recipient's email.	System accepts the recipient identifier.
2	User enters the amount to send.	System checks wallet balance (<<include>> Check Wallet Balance).
3		System deducts the amount from the wallet, records the transaction as "COMPLETED", and updates the balance file.
4		System displays a success confirmation.

Alternate Courses:	If the wallet balance is insufficient, an InsufficientFundsException is thrown and an error is displayed.
Implementation Constraints:	Amount must be greater than \$0.00.

Use Case 4: Request Money

Use Case Name	Request Money
Actor(s):	User
Description:	Allows the user to send a payment request to another person.
Precondition:	User is logged in.

Step #	Actor	System
1	User selects "Request Money" and enters the other person's email.	System accepts the identifier.
2	User enters the requested amount.	System creates a TransactionRecord with status "PENDING" and saves it to file.
3		System displays a confirmation that the request was sent.

Alternate Courses:	No funds are transferred at this stage. The request remains PENDING.
Implementation Constraints:	Amount must be greater than \$0.00.

Use Case 5: Add Funds to Wallet

Use Case Name	Add Funds to Wallet
Actor(s):	User, Bank System
Description:	Allows the user to deposit funds from a linked payment method into their wallet balance.
Precondition:	User is logged in and has at least one linked payment method.

Step #	Actor	System
1	User selects "Add Money to Wallet" and chooses Bank Account or Credit Card.	System displays available payment methods of the selected type.
2	User selects a payment method and enters the amount.	System submits the request to the payment method (<<include>> Process Deposit).
3	Bank Account is debited or Credit Card is charged.	
4		System credits the wallet, records the transaction as "COMPLETED", and updates the balance file.

Alternate Courses:	If the bank account has insufficient balance, an InsufficientFundsException is thrown and no balance change occurs.
Implementation Constraints:	Amount must be greater than \$0.00. User must have at least one payment method linked.

Use Case 6: Withdraw to Bank

Use Case Name	Withdraw to Bank
Actor(s):	User, Bank System
Description:	Allows the user to transfer funds from their wallet to a linked bank account.
Precondition:	User is logged in, has sufficient wallet balance, and has a linked bank account.

Step #	Actor	System
1	User selects "Withdraw to Bank" and selects a linked bank account.	System displays available bank accounts.
2	User enters the withdrawal amount.	System checks wallet balance (<<include>> Check Wallet Balance).
3		System submits the request to the Bank System (<<include>> Process Withdrawal). Bank credits the account.
4		System deducts from wallet, records transaction as "COMPLETED", and updates the balance file.

Alternate Courses:	If the wallet balance is insufficient, an InsufficientFundsException is thrown and no transfer occurs.
Implementation Constraints:	Only bank accounts are valid withdrawal targets. Amount must be greater than \$0.00.

Use Case 7: View Transaction History

Use Case Name	View Transaction History
Actor(s):	User
Description:	Allows the user to view all past transactions associated with their wallet.
Precondition:	User is logged in.

Step #	Actor	System
1	User selects "Transaction History".	System retrieves all TransactionRecord objects for the current user.
2		System displays each record showing: ID, type, amount, counterparty, and status.

Alternate Courses:	If no transactions exist, the system displays "No transactions yet."
Implementation Constraints:	Only transactions belonging to the authenticated user are displayed.

Use Case 8: Link Payment Method

Use Case Name	Link Payment Method
Actor(s):	User
Description:	Allows the user to link a bank account or credit card to their wallet profile.
Precondition:	User is logged in.

Step #	Actor	System
1	User selects "Link Payment Method" and chooses Bank Account or Credit Card.	System displays the appropriate input form.
2	User fills in the required details (holder name, account/card number, etc.).	System validates inputs are not empty.
3		System adds the payment method to the user's profile and saves it to file.

Alternate Courses:	If a duplicate payment method is detected (same last 4 digits), the system does not add it again.
Implementation Constraints:	Holder name and account/card number are required. Credit cards require expiry date and card type.